TRUST-Campus: Evidence-Aware Retrieval-Augmented Generation for Reliable University Information Access

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Abstract—Retrieval-augmented generation (RAG) can surface relevant policy documents while still producing unreliable answers when retrieved evidence is incomplete, user-specific context is missing, or a query contains a misleading premise. This paper presents TRUST-Campus, an evidence-aware RAG framework for university information access that explicitly separates retrieval relevance from answer sufficiency. The system retrieves candidate evidence and applies a semantic sufficiency decision layer before generation, allowing it to answer when the available evidence supports the requested conclusion and abstain otherwise. We evaluate Dense RAG, Hybrid RAG, Reranked RAG, and Evidence-Aware RAG V4 on a frozen 50-question evaluation set covering paraphrase, compositional reasoning, negation, context-dependent questions, unsupported requests, future questions, and false premises. All four systems achieved a document retrieval hit rate of 1.000, isolating answer-sufficiency reasoning as the primary source of performance differences. Evidence-Aware V4 achieved the highest answer/abstain decision accuracy (0.820), highest coverage (0.420), and lowest false rejection rate (0.200), compared with 0.740 decision accuracy, 0.180 coverage, and 0.600 false rejection rate for the strongest conventional baseline. Reranked RAG remained more conservative, achieving the lowest false acceptance rate (0.033 versus 0.167 for V4). Category-level analysis shows that evidence-aware sufficiency reasoning substantially improves paraphrase, compositional, and negation cases, while personalized context-dependent questions remain the dominant failure mode. These findings support a broader design principle for trustworthy RAG: retrieving relevant evidence is not enough; a system should determine whether the retrieved evidence is sufficient to justify answering the specific question.

Keywords—retrieval-augmented generation, evidence sufficiency, abstention, trustworthy AI, question answering, university policy, selective prediction

1. Introduction

Retrieval-augmented generation (RAG) combines language generation with external evidence retrieval, offering a practical way to ground answers in source documents rather than relying only on parametric model knowledge [1]. Dense retrievers and reranking pipelines can substantially improve access to relevant passages [2], while more recent systems explore adaptive retrieval and self-reflection [3], [4]. Yet retrieval quality and answer reliability are not equivalent. A system can retrieve the correct policy document and still answer a question that the document does not fully resolve.

This distinction is particularly important in university information systems. Questions about course withdrawal, experiential learning, co-op participation, and employment authorization often appear superficially similar while differing in decisive details such as program status, timing, employer, work location, immigration status, or current enrollment. A retrieved passage may be highly relevant to the topic but insufficient to determine what is true for a particular student. Conversely, a conservative system may abstain even when multiple retrieved statements jointly support a clear answer.

TRUST-Campus is designed around the hypothesis that trustworthy RAG should make an explicit evidence-sufficiency decision between retrieval and generation. Rather than asking only whether retrieved passages are relevant, the system asks whether the available evidence is sufficient to justify the specific answer requested. This framing connects RAG with selective prediction, where a model may reject an input rather than make a low-confidence decision [5].

The project makes three contributions. First, it introduces an evidence-aware RAG pipeline that treats answerability as a semantic sufficiency problem rather than a retrieval-score threshold alone. Second, it defines a controlled

benchmark that separates seven question types designed to expose over-answering and over-abstention failure modes. Third, it compares evidence-aware RAG with dense, hybrid, and reranked RAG under a shared frozen evaluation protocol and analyzes the resulting coverage-risk tradeoff.

2. Related Work

2.1 Retrieval-Augmented Generation

Lewis et al. introduced RAG as a framework that combines parametric sequence-to-sequence models with non-parametric retrieved memory for knowledge-intensive generation [1]. Dense Passage Retrieval (DPR) demonstrated that learned dense representations can provide strong retrieval performance for open-domain question answering [2]. These approaches established retrieval as an effective mechanism for grounding model outputs in external evidence.

However, standard retrieve-then-generate pipelines generally assume that once relevant documents have been retrieved, generation should proceed. This assumption is problematic when retrieved evidence is topically related but incomplete for the specific decision being requested. TRUST-Campus focuses on this gap between relevance and sufficiency.

2.2 Adaptive Retrieval and Self-Reflection

Later work has explored adaptive retrieval. FLARE retrieves additional information during generation when upcoming content appears uncertain [3]. Self-RAG trains a model to retrieve and critique evidence and its own generations using learned reflection signals [4]. These methods demonstrate that retrieval need not be a fixed one-time operation. TRUST-Campus is complementary: it concentrates on whether already retrieved evidence is sufficient to support answering at all, particularly in policy-oriented question answering where abstention may be preferable to an unsupported personalized conclusion.

2.3 Selective Prediction and Abstention

Selective prediction studies the tradeoff between coverage and risk by allowing a model to abstain on uncertain instances [5]. This perspective is useful for RAG because an answer can be harmful even when it is fluent and based on a relevant passage. TRUST-Campus therefore treats answer/abstain behavior as a first-class evaluation target and reports coverage, false rejection, false acceptance, and decision accuracy in addition to retrieval and answer-overlap metrics.

3. Problem Formulation

Let q denote a user question and R(q) the set of evidence passages returned by a retriever. Conventional RAG primarily optimizes the relevance of R(q) and then conditions a generator on those passages. TRUST-Campus inserts a sufficiency decision S(q, R(q)) before generation. The system answers only when the retrieved evidence is judged sufficient to resolve the question; otherwise it abstains with an explicit insufficient-evidence response.

The central distinction is: relevant evidence concerns the same topic, whereas sufficient evidence supports the specific conclusion requested by the query. Evidence can be relevant but insufficient when the user asks for a personalized determination without providing required state, asks for a future fact not present in the corpus, asks for an unsupported statistic, or embeds a false premise that cannot safely be accepted.

4. Methodology

4.1 Corpus and Task Domain

The prototype uses a curated university-information corpus spanning three policy domains: course drop/withdrawal rules, graduate co-op policy, and curricular practical training (CPT) guidance. These domains were selected because they contain factual policy statements as well as questions whose answerability depends on timing, student-specific context, or combinations of multiple rules.

4.2 Compared Systems

We evaluate four systems under the same benchmark and metric definitions:

- Dense RAG: a standard dense-retrieval pipeline followed by answer generation.
- Hybrid RAG: a retrieval pipeline that combines complementary retrieval signals before generation.
- Reranked RAG: a retrieval pipeline that reranks candidate evidence prior to answer generation.
- Evidence-Aware RAG V4: the proposed system, which applies semantic evidence-sufficiency logic after retrieval and before answering.

The final Evidence-Aware V4 implementation was frozen before the final evaluation. The evaluator imported the active V4 module, and the active RAG and semantic-sufficiency source files were verified byte-for-byte identical to their frozen copies before paper finalization.

4.3 Evidence-Sufficiency Decision Layer

The sufficiency layer is designed to reject evidence that is merely related to the query but does not resolve the requested decision. In conceptual terms, it checks whether the retrieved evidence contains the conditions, scope, and policy relationships needed to support an answer. The decision policy is intentionally conservative for unsupported, future, and context-dependent questions, while allowing answer generation when the evidence can support a direct, compositional, or negated conclusion.

This layer is not presented as a learned universal verifier. It is a task-specific semantic decision mechanism developed for the TRUST-Campus setting. The evaluation therefore asks whether this explicit intermediate decision improves answerability behavior relative to conventional retrieval pipelines, not whether it solves evidence verification in general.

4.4 Frozen Final Evaluation Set

The final evaluation set contains 50 questions. It was frozen before the reported final comparison and no V4 tuning was performed after observing these results. The set contains 20 answerable and 30 unanswerable questions, stratified across seven categories. Because the benchmark was constructed within this project rather than collected independently from an external source, we refer to it as a frozen final evaluation set rather than an external held-out benchmark.

Category	Count	Answerability focus
Paraphrase	8	Answerable
Compositional	6	Answerable
Negation	6	Answerable
Needs context	10	Unanswerable
Unsupported	10	Unanswerable
Future	5	Unanswerable
False premise	5	Unanswerable

Table 1. Composition of the frozen final evaluation set.

4.5 Metrics

Document Retrieval Hit Rate measures whether at least one gold supporting document is present in the retrieved evidence for questions that have gold documents. Coverage is the fraction of all questions for which the system returns an answer rather than abstaining. Answer/Abstain Decision Accuracy is the fraction of questions for which the system makes the correct answerability decision. False Rejection Rate (FRR) is the fraction of answerable questions incorrectly rejected. False Acceptance Rate (FAR) is the fraction of unanswerable questions incorrectly answered. Token F1 measures lexical overlap between the generated answer and gold answer on answered, answerable questions.

These metrics intentionally separate answerability from answer wording. A system may have high token F1 on the small subset it chooses to answer while still being excessively conservative. Conversely, a system may improve coverage and decision accuracy while producing answers with lower lexical overlap. The evaluation therefore treats token F1 as complementary rather than as the primary reliability metric.

5. Results

5.1 Overall Performance

System	Retrieval Hit	Coverage	Decision Acc.	FRR	FAR	Token F1
Dense RAG	1.000	0.220	0.660	0.650	0.133	0.507
Hybrid RAG	1.000	0.260	0.660	0.600	0.167	0.480
Reranked RAG	1.000	0.180	0.740	0.600	0.033	0.523
Evidence-Aware V4	1.000	0.420	0.820	0.200	0.167	0.443

Table 2. Final comparison on the frozen 50-question evaluation set.

All four systems achieved a document retrieval hit rate of 1.000. The final comparison therefore isolates post-retrieval answerability behavior rather than retrieval failure. Evidence-Aware V4 achieved the highest overall decision accuracy at 0.820, compared with 0.740 for Reranked RAG and 0.660 for both Dense and Hybrid RAG. It also achieved the highest coverage at 0.420 and the lowest false rejection rate at 0.200.

Compared with Reranked RAG, the strongest conventional baseline on decision accuracy, V4 improved decision accuracy by 8 percentage points, increased coverage by 24 percentage points, and reduced false rejection from 0.600 to 0.200. This improvement came with a higher false acceptance rate: 0.167 for V4 compared with 0.033 for Reranked RAG. Accordingly, V4 should be interpreted as improving the coverage-correctness tradeoff rather than being uniformly more conservative.

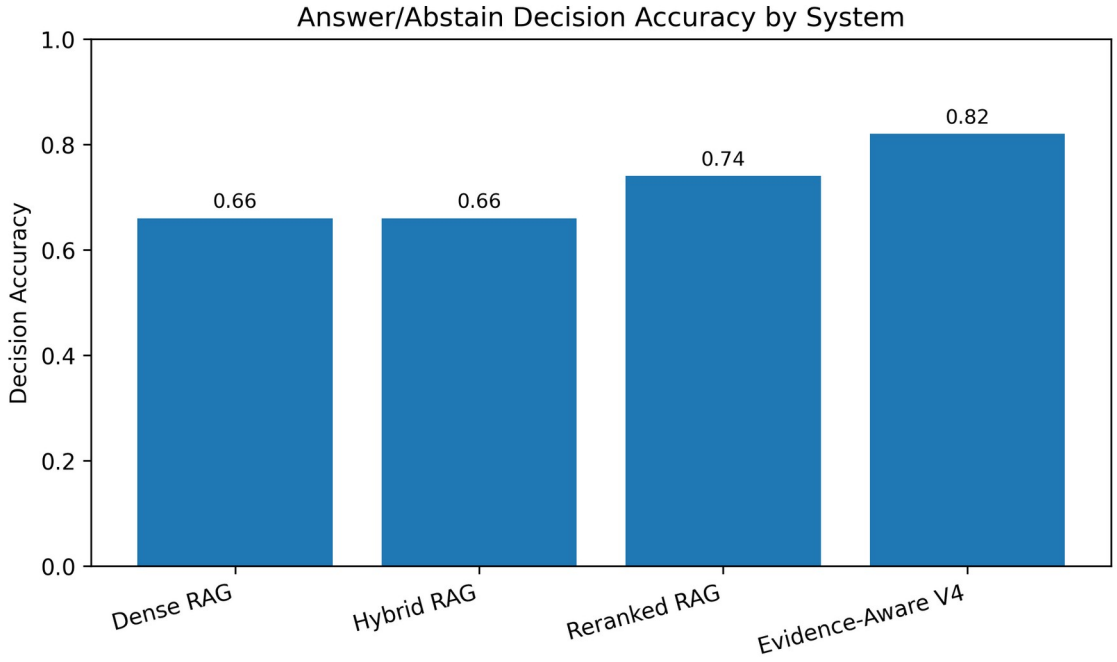


Figure 1. Answer/abstain decision accuracy by system.

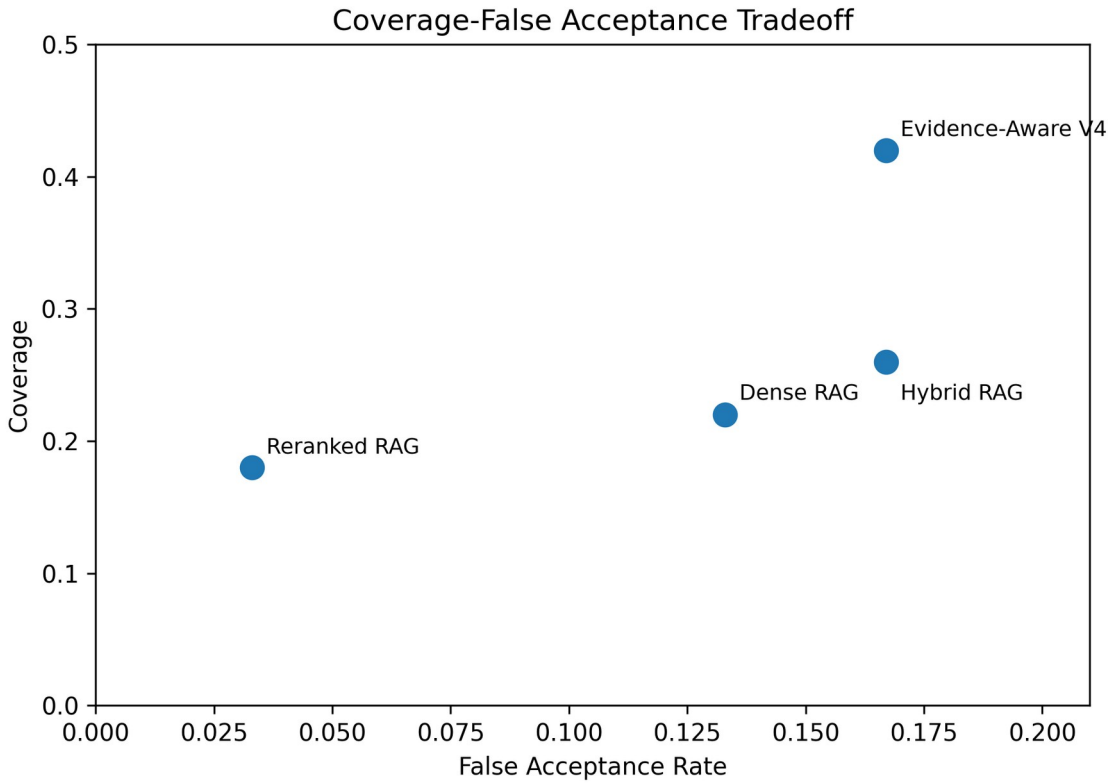


Figure 2. Coverage versus false acceptance rate. Higher coverage is preferable, while lower false acceptance is safer.

5.2 Performance by Question Category

Category	Dense	Hybrid	Reranked	Evidence-Aware V4
Paraphrase	0.750	0.750	0.750	1.000
Compositional	0.000	0.167	0.333	0.667
Negation	0.167	0.167	0.000	0.667
Needs context	0.600	0.500	0.900	0.600
Unsupported	1.000	1.000	1.000	1.000
Future	1.000	1.000	1.000	1.000
False premise	1.000	1.000	1.000	0.800

Table 3. Decision accuracy by question category.

Evidence-Aware V4 achieved perfect accuracy on paraphrase questions and substantially outperformed all three baselines on compositional and negation questions. On compositional questions, V4 reached 0.667 accuracy, compared with 0.333 for Reranked RAG, 0.167 for Hybrid RAG, and 0.000 for Dense RAG. On negation questions, V4 again reached 0.667, while the baselines ranged from 0.000 to 0.167.

All four systems achieved perfect decision accuracy on unsupported and future-oriented questions, suggesting that explicit absence of supporting evidence was comparatively easy to recognize. The most difficult category for V4 was needs-context, where it achieved 0.600 accuracy and produced four false acceptances. Reranked RAG was substantially more conservative in this category, reaching 0.900 accuracy with only one false acceptance. V4 also made one false acceptance in the false-premise category.

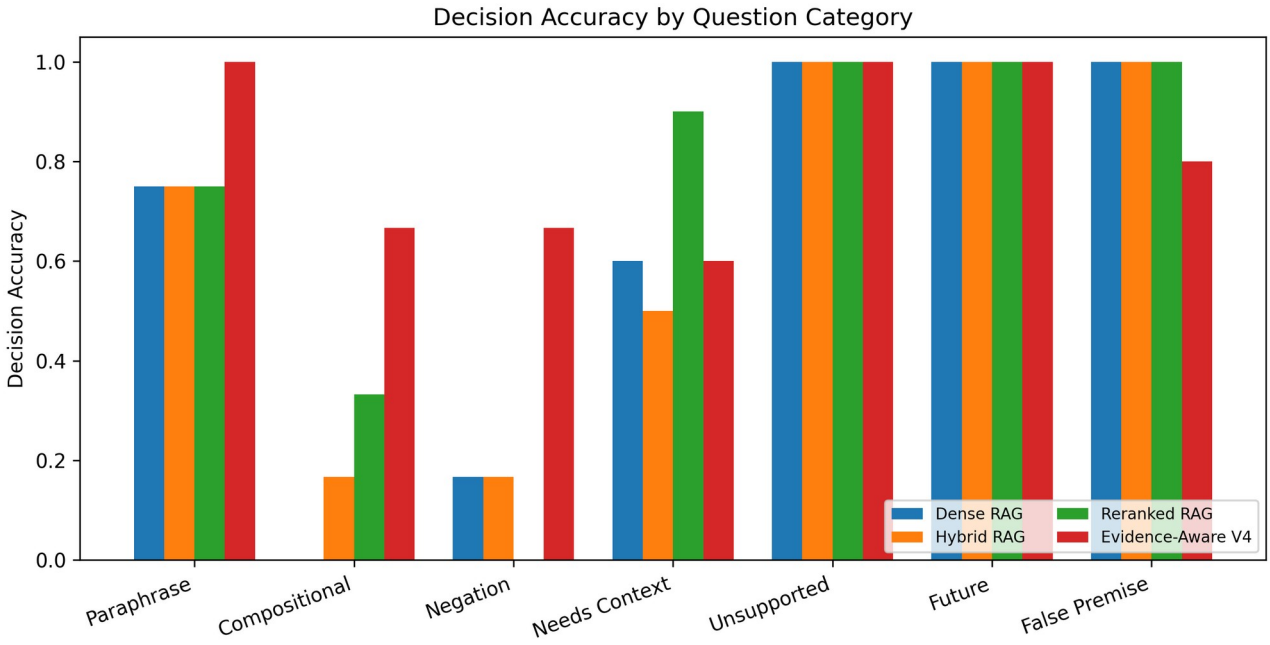


Figure 3. Decision accuracy by question category.

6. Error Analysis

6.1 False Rejections

Evidence-Aware V4 falsely rejected four answerable questions: f009, f011, f019, and f020. Dense RAG falsely rejected 13 answerable questions, while Hybrid and Reranked RAG each falsely rejected 12. The remaining V4 false rejections are concentrated in compositional and negated policy questions. These cases require the system to infer that retrieved rules jointly support a negative conclusion rather than matching a directly stated affirmative answer.

This pattern shows that evidence sufficiency is not reducible to retrieval confidence. The correct source document may be present, yet the system must still recognize logical scope, combine conditions, or interpret negation correctly. The large reduction in false rejections suggests that semantic sufficiency reasoning helps, but the residual errors show that composition remains incomplete.

6.2 False Acceptances

Evidence-Aware V4 falsely answered five unanswerable questions: f024, f026, f028, f030, and f049. Four were needs-context questions and one was a false-premise question. These errors arise when a general policy statement appears sufficient at the topic level but does not contain enough information to make a determination about the individual user.

For example, a policy may state a general co-op duration or withdrawal rule, but a question phrased as “How long am I allowed to stay?” or “Can I still drop my course?” can depend on information not contained in the query or corpus context. This exposes a central distinction between policy relevance and personalized decision sufficiency.

6.3 Retrieval Is Necessary but Not Sufficient

Because every system achieved perfect document-level retrieval hit rate on items with gold supporting documents, the observed errors are predominantly post-retrieval errors. Improving the retriever alone would not resolve most of them. Reliable RAG therefore requires an explicit mechanism for deciding whether the evidence actually resolves the user’s question.

7. Discussion

The results support three primary observations. First, explicit evidence-sufficiency reasoning can substantially reduce unnecessary abstention. Evidence-Aware V4 reduced false rejection from 0.600 for the strongest conventional baseline to 0.200 while increasing coverage from 0.180 to 0.420. Second, retrieval sophistication alone does not guarantee better answerability decisions. Reranked RAG achieved the lowest false acceptance rate and the highest token F1, but it remained highly conservative and rejected 60% of answerable questions. Third, context-dependent questions remain the hardest unresolved case for the evidence-aware system.

The category analysis is particularly revealing. V4's gains on paraphrase, compositional reasoning, and negation suggest that an explicit semantic sufficiency layer can recover answerable cases that conventional pipelines incorrectly reject. At the same time, V4's errors on needs-context queries show that a system can overgeneralize from a correct policy passage when the user-specific variables required for a determination are missing.

This motivates a broader design principle: RAG systems should separate three questions that are often conflated—Was relevant evidence retrieved? Does that evidence logically support a conclusion? Is the available information sufficient for the specific user-level decision being requested? TRUST-Campus addresses the second and third questions explicitly, but the remaining false acceptances show that personalized sufficiency deserves further work.

The comparison also demonstrates that there is no single universally optimal operating point. Reranked RAG offers a low false acceptance rate but very low coverage. Evidence-Aware V4 answers more useful questions and achieves higher overall decision accuracy, but at the cost of more false acceptances. The preferred operating point should therefore depend on deployment risk. In high-stakes advising, a lower FAR threshold may be appropriate; in lower-risk information access, higher coverage may be preferable.

8. Threats to Validity and Limitations

The evaluation has several important limitations. First, the frozen final set contains only 50 questions and covers a constrained university-policy domain. The results should not be generalized to arbitrary RAG applications without additional evaluation. Second, the benchmark was constructed within the project and was not independently collected from an external population; accordingly, it is described as a frozen final evaluation set rather than an external held-out benchmark.

Third, all four systems achieved perfect document retrieval hit rate on the answerable items in this evaluation. This is useful for isolating evidence-sufficiency behavior but means the study does not characterize robustness under severe retrieval failure. Fourth, category sizes are small: compositional and negation contain six questions each, and false premise contains five. A single error can therefore substantially change category-level accuracy.

Fifth, token F1 should not be interpreted as a complete measure of answer quality. V4 has lower token F1 than Reranked RAG despite higher decision accuracy because token F1 measures lexical overlap only on answered, answerable questions. Finally, the semantic sufficiency mechanism is task-specific and partly rule-informed; broader generalization would require evaluation across institutions, policy types, retrievers, models, and naturally occurring user queries.

9. Future Work

The most immediate direction is explicit missing-context modeling. Instead of only answering or abstaining, a future system could identify the missing variable—for example, program status, authorization status, course timing, employer, or work location—and ask a targeted clarification question. This would transform some current abstentions and false acceptances into interactive evidence gathering.

A second direction is calibrated sufficiency scoring. The current evidence-aware mechanism can be extended into a probabilistic decision model whose threshold is tuned to a target false acceptance rate, enabling deployment-specific risk control. A third direction is contradiction and premise checking: queries that embed false assumptions should be decomposed so the system can explicitly reject the premise before considering the requested answer.

Finally, future evaluation should include larger independently collected test sets, naturally occurring advising questions, cross-institution transfer, retrieval corruption tests, and human evaluation of answer correctness,

helpfulness, and abstention quality. These additions would clarify whether the observed advantages persist beyond the controlled benchmark.

10. Conclusion

TRUST-Campus investigated whether a RAG system can distinguish evidence that is merely relevant from evidence that is sufficient to justify an answer. On a frozen 50-question university-information evaluation set, Evidence-Aware RAG V4 achieved 0.820 answer/abstain decision accuracy, compared with 0.740 for Reranked RAG and 0.660 for Dense and Hybrid RAG. It increased coverage to 0.420 and reduced false rejection to 0.200, compared with 0.180 coverage and 0.600 false rejection for the strongest conventional baseline.

The largest improvements occurred on paraphrased, compositional, and negated questions, indicating that evidence-sufficiency reasoning can help recover answerable cases that conventional pipelines reject. The remaining weaknesses are concentrated in context-dependent questions and one false-premise case, showing that general policy relevance can still be mistaken for personalized sufficiency.

The core conclusion is therefore simple: trustworthy RAG should not stop at retrieving relevant evidence. It should determine whether the available evidence is sufficient to support the specific action of answering.

References

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Appendix A. Final Evaluation Summary

The final benchmark contains 20 answerable and 30 unanswerable questions. Evidence-Aware V4 produced four false rejections (f009, f011, f019, f020) and five false acceptances (f024, f026, f028, f030, f049). The corresponding category-level decision accuracies were 1.000 for paraphrase, 0.667 for compositional, 0.667 for negation, 0.600 for needs-context, 1.000 for unsupported, 1.000 for future, and 0.800 for false-premise questions.

For reproducibility, the final V4 evaluation was run using an evaluator that imports the active V4 RAG implementation. The active V4 RAG file and semantic-sufficiency file were subsequently verified byte-for-byte identical to their frozen copies. The three baseline outputs and the V4 output were preserved as separate JSONL result files, and the project configuration was restored after baseline evaluation.